House Wiring Rules as Per ISI Specification

In construction of a house, the owner should focus on electric works, which need a lot of attention while laying wiring as well as quality of wiring, specification of electric appliances and cables/wires. Always plan to work exactly where the outlets, switches and fixtures are going to be placed. This also allows to check the appropriate tools and materials are in sufficient quality. Calculate the total load from various items that would be used in the house and select the proper conductor that is capable of withholding the total load and distribute accordingly.

Care should be taken while fitting pipes, junction boxes, fan hook boxes in RCC slab and walls. If any fault happens, it may cause any incident and may require dismantling of the defective portion. So take proper care to avoid dismantling which is very costly in comparison to get them fixed properly earlier. To avoid an electric shock, the following precautions must be kept in mind.Therefore the following rules laid down by the Indian Standard Institution should be followed. Few of them are listed below for your guidance.

- The wiring should be carried out on distribution systems with main and branch distribution boards.
- All conductors should run along walls and ceiling, so that they are easily accessible and capable of being thoroughly inspected. In any case, wiring should not be run above ceiling.
- Horizontal run of wiring should be at a height of 3 metre.Switch Boards should be fitted at a height of 1.5 metre.
- Fuse wire should be connected with phase only. Connect all switches with phase wire. Connect the neutral link in neutral wire.
- One circuit means, one connection from electrical meter or main fuse board.The number of points in light circuit should not exceed 10 (or) total load on circuit should not exceed 800 Watts.
- All conductors should be made of copper and should be stranded. They should have a cross section less than 0.002 square inches, nominal area (3/0.029 inches).
- For Power wiring circuit, the size of wire should be 1.5 mm square for copper and 2 mm square for aluminium.[One power circuit = (three 5 amp socket) or (two 15 amp socket) or (one 15 amp socket + two 5 amp socket) or (one Alternating current circuit)]
- Never use damaged insulation, for wiring. It avoid short circuit and overloading, with the use of MCB and save the electrical appliances.
- Burnt element, cut/broken wiring, loose/open connection should be avoided.
- Earthing means to connect electrical system to general mass of earth to ensure immediate discharge of electrical energy without danger. Provide earth connection to enhance voltage and protect human beings from sudden electric shock.Earth wire should be 14 SWG in case of copper and 4 mm square in case of Aluminium.

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